1. PROTOCOL

RNA purification, reverse transcription and PCR amplification are carried out in dedicated facilities. AMPLIFICATION OF PR-RT REGION

- *VIRAL RNA PURIFICATION*
- Use NHP (normal human plasma) or, if not available, buffer as a negative control to rule out contamination.
- Before working with RNA, clean lab bench and pipettes with an RNase decontamination solution (e.g., RNase AWAY™ Decontamination Reagent).

  - *RNA EXTRACTION WITH PURELINK® VIRAL RNA/DNA MINI KIT*

- When using the kit for the 1^{st} time, prepare Carrier RNA (5.6 µg/sample). Add 310 µL RNase-free Water (included with the kit) to 310 µg lyophilized Carrier RNA supplied in a tube to obtain 1 µg/µL Carrier RNA stock solution. Mix thoroughly and aliquot the solution into smaller aliquots. Store aliquots at –20°C; avoid repeated freezing and thawing.
- Calculate the volume of Lysis Buffer/Carrier RNA mix required to process the desired number of samples simultaneously according to the table below.

Combine the following:	**Per run**	**6 runs**	**10 runs**	**24 runs**
Lysis Buffer	210 µL	1260 µL	2100 µL	5040 µL
Carrier RNA	6 µL	35 µL	59 µL	141 µL

Thaw the required amount of 1 µg/µL Carrier RNA stock solution. In a sterile tube, add the volume of Carrier RNA to the Lysis Buffer. Mix gently by pipetting up and down. Avoid vortexing as this generates foam.

1. Add 25 µL Proteinase K (included with the kit) to a sterile microcentrifuge tube.
2. Add 200 µL of cell-free sample (equilibrated to room temperature) to the microcentrifuge tube. **Note**: If <200 µL sample are processed, adjust final volume of the sample to 200 µL using PBS or 0.9% NaCl.
3. Add 200 µL Lysis Buffer (containing 5.6 µg Carrier RNA). Close lid and mix by vortexing for 15 seconds.
4. Incubate at 56°C for 15 minutes.
5. Briefly centrifuge tube to remove any droplets from the inside of the lid.
6. Add 250 µL 96–100% ethanol to the lysate tube to obtain a final ethanol concentration of 37%, close lid, mix by vortexing for 15 seconds. **Note**: If processing up to 10 samples, ethanol may be added to all tubes prior to vortexing.
7. Incubate lysate with ethanol for 5 minutes at room temperature.
8. Briefly centrifuge tube to spin down any droplets from lid.
9. Transfer above lysate with ethanol (~675 µL) onto the Viral Spin Column.
10. Close lid and centrifuge column at 6,800 x g for 1 minute. Discard collection tube with the flowthrough.
11. Place spin column in a clean Wash Tube (2 mL) included with the kit and add 500 µL Wash Buffer (WII) with ethanol to spin column.
12. Close lid and centrifuge the column at 6,800 x g for 1 minute. Discard flow-through and place spin column back into the Wash Tube.
13. Again add 500 µL Wash Buffer (WII) with ethanol to spin column.

14. Close lid, centrifuge at 6,800 x g for 1 minute. Discard Wash Tube containing flow-through.
15. Transfer spin column to a clean Wash Tube (2 mL) included with the kit.
16. Centrifuge column at maximum speed in a microcentrifuge for 1 minute to completely dry the membrane. Discard Wash Tube with the flowthrough.
17. Place Viral Spin Column in a clean 1.5 mL Recovery Tube supplied with the kit.
18. Add 50 µL of sterile, RNase-free water (E3) to center of the column. Close lid.
19. Incubate at room temperature for 1 minute.
20. Centrifuge column at maximum speed for 1 minute. The Recovery Tube contains now purified viral nucleic acids. Remove and discard spin column.
21. Store the purified RNA/DNA at –80°C or use RNA/DNA immediately for the intended downstream application.

REVERSE TRANSCRIPTION LINKED TO FIRST PCR

Care should be exercised when handling RNA, which is very labile. Work diligently and rapidly. Always wear clean gloves. Keep RNA samples and mixes until PCR reaction.

1. In **Pre PCR area**, in **PCR hood**, dispense 1 µL of primer "R-RT-NcoI4363" at 10 µM in each PCR tube
2. Add 5 µL of RNA sample
3. Start One-Step RT-PCR reaction using program RNA 65 on the GeneAmp® 9700 cycler (volume 6 µL); 5 minutes at 65°C
4. Transfer primer/RNA mix in the purple cooling box and chill for 2 minutes.
5. In the meantime, prepare a master mix containing (n+1 reactions):

   USE Document RT-PCR-MM for Calculation; take note of primer concentrations!!

   1 µL primer "F-1818" at 10 µM

   25 µL Herculase II RT-PCR 2x Master Mix

   16 µL H_2O

   1 µL Affinity Script RT/RNase Block **(keep refrigerated at all times!)**

6. Keep master mix in cooling box. Dispense 44 µL of the reaction mix to each PCR tube
7. Start one-step RT-PCR using program PR-RT-PCR1 on the GeneAmp® 9700 cycler in **main lab** (volume 50 µL):

   30 minutes at 45°C

   1 minute at 92°C

   20 seconds at 92°C

   20 seconds at 55°C 40 cycles

   3 minutes at 68°C

   5 minutes at 68°C

   Pause at 4°C

EXOPROSTAR™ TREATMENT

1. work strictly **ON ICE.** In **Post PCR area**, add 5 µL of ExoProStar™ to each first PCR sample (prior to use, mix content of the ExoProStar™ solution by flicking the closed tube several times).
2. Digest primers using program Exostar on the GeneAmp®9700 cycler in **main lab** (50 µL):

    15 minutes at 37°C

    15 minutes at 80°C

    Pause at 4°C

SECOND PCR AMPLIFICATION

In **Pre PCR area**, in **PCR hood**, prepare a reaction mix for n+1 reactions by mixing the following:

1.5 µL primer F-2005 at 10 µM

1.5 µL primer R-4303 at 10 µM

10 µL 5x KAPA Fidelity Buffer

1.5 µL 10 mM KAPA dNTPs

1 µL KAPA Polymerase Hotstart **(to be kept refrigerated at all times!)**

29.5 µL H2O

Keep reaction in purple cooling box and avoid warming to room temperature after addition of KAPA and primers. Dispense 45 µL of the reaction mix in n fresh PCR tubes **in purple cooling box**

3. In **Post PCR area**, add 5 µL of each first round PCR product
4. Start 2nd PCR using program PR-RT-PCR2 on the GeneAmp® 9700 cycler in **main lab**:

    3 minutes at 95°C

    20 seconds at 98°C

    15 seconds at 60°C 35 cycles

    2 minutes at 72°C

    5 minutes at 72°C

    Pause at 4°C

GEL PURIFICATION OF THE AMPLICONS

PREPARING THE GEL

NOTE: Ethidium bromide is toxic. Always wear clean nitrile gloves. Use it only in dedicated area and discard it in the respective bin.

Weigh desired amount of agarose in a glass bottle; mix with appropriate volume of 1X TBE buffer according to the required gel percentage (i.e. for 1% agarose gel, mix 1g of agarose with 100 mL TBE buffer). For one full-size gel, 400 mL are needed, and 100mL for full gel cassettes in small tank.

- Boil mixture until agarose is completely dissolved (Note: cap must be untightened!). Let bottle cool at room temperature.
- When agarose temperature is <50°C (hold in your hand), add 7 µL Ethidium Bromide solution per 100 mL agarose gel. (If too hot, ethidium bromide may be degraded.)
- Mix well and cast gel. Remove bubbles with a tip if needed. Let solidify at room temperature.

Loading and Running the Gel

1. Add **10 µL** of 6x Crystal Violet loading dye to 50 µL of DNA sample and load **app. 20 µL** onto gel.
2. Run gel at 35V for 10 minutes and then at 50V for 20 minutes. A thin violet band should be visible, moving down into the gel. If no band is visible, insufficient DNA was loaded (< 200 ng).
3. If DNA bands is sufficiently separated so that the fragment is easily excised, switch power supply off. Proceed directly to Excision of DNA. **The correct fragment size is ca. 2300 bp. If the main fragment is clearly below or above that size, do not use for purification** but repeat PCR.

Excising DNA

1. Transfer the gel onto saran wrap.
3. Using a new scalpel, carefully excise the correct DNA fragment from the gel.
4. Transfer the excised agarose plug to a sterile 1.5 ml microcentrifuge tube.

Purifying DNA Fragment

1. Weigh gel slice.
2. For each 100 mg of agarose gel add 200 µL Buffer NTI (e.g. for 200mg of agarose add 400 µL Buffer NTI, for 300mg of agarose add 600 µL Buffer NTI).
3. Incubate sample for 10 minutes at 50°C using Thermomixer. Vortex the sample briefly every 2 – 3 minutes until the gel slice is **completely** dissolved.
4. Place a NucleoSpin® Gel and PCR Clean-up Column into a Collection Tube and load up to 700 µL sample.
5. Centrifuge for 1 minute at 11,000 x g. Discard flow-through and place column back into collection tube. Load remaining sample **if necessary** and centrifuge for 1 minute at 11,000 x g.
6. Add 700 µL Wash Buffer NT3 to the NucleoSpin® Gel and PCR Clean-up Column. Centrifuge for 1 minute at 11,000 x g. Discard flow-through and place column back into the collection tube.
7. For a second time, add 700 µL Wash Buffer NT3 to the NucleoSpin® Gel and PCR Clean-up Column. Centrifuge for 1 minute at 11,000 g. Discard flow-through and place the column back into the empty collection tube.
8. Centrifuge empty tubes for 2 minutes at 11,000 x g to remove residual Wash Buffer NT3. **Make sure spin column does not come in contact with flow-through while removing it from collection tube.**
9. Place NucleoSpin® Gel and PCR Clean-up Column into a new 1.5mL microcentrifuge tube.
10. Add 40 µL Buffer NE and incubate at room temperature **for 1 minute**.
11. Centrifuge for 2 minutes at 11,000 x g.

SEQUENCING OF THE AMPLICONS

SEQUENCING REACTION

Defreeze primers and predilute in pre-PCR room: Predilution: 1/100 -> 1µL Primer, 99µL H_2O

In **main lab**, prepare a reaction mix for n+1 reactions by mixing the following:

3 µL F-2005 or F-2135 or F-2610 or R-2605 or R-3010 or R-3584 primer at 1 µM

1 µL BrightDye Terminator Sequencing kit

3 µL 5X Sequencing Buffer

X µL H_2O (up to 20 µL)

Dispense Y µL of the reaction mix in n fresh PCR tubes; add Z µL of each sample

Calculation for 1 well containing 1 primer – remember that you need 6 wells for 1 patient!!

Mastermix	**Primer**	**DNA**
1 µL BrightDye Terminator 3 µL 5X Sequencing Buffer X µL H_2O (up to 20 µL)	3 µL Primer (F-2005 **or** F-2135 **or** F-2610 **or** R-2605 **or** R-3010 **or** R-3584)	Z µL of DNA sample To reach 40-100 ng, e.g. 1 µL of sample that contains 70 ng/µL or 2 µL of sample that contains 30 ng/µL

Total volume of MM is volume Y

Total volume of this makes 20 µL

- Start sequencing reaction using program SEQ.REAK on the GeneAmp® 9700 cycler in **main lab**:

20 seconds at 96°C
20 seconds at 50°C
4 minutes at 60°C

40 cycles

Pause at 4°C

PURIFICATION OF SEQUENCING REACTION

1. Prepare Sephadex 96-well plate for purification of sequencing reactions by filling the black metal plate with Sephadex powder. If less than a whole plate is needed, cover neighboring cavities with tape. Fill with spatula and tap on bench to compact Sephadex well.
2. Place Sephadex plate (with filter at the bottom) on top of the black metal plate and flip the array over. Tap it on bench to transfer all Sephadex powder to the plate.
3. Add 300 µL of deionised, DNase-free water to each well without touching the Sephadex; let swell for 3-4 hours at room temperature (or 30 min at 37°C).
4. With the blue rubber adapter, fix Sephadex plate to top of the collection plate. (use another (used) Sephadex plate for counterbalancing; adjust weight on lab balance)

   - Centrifuge 5 minutes at 600 x g.

- Discard water and store collection plates.

5. Fix the Sephadex plate on the top of the sequencing plate.

   - Apply the sequencing reaction samples onto the Sephadex without touching it.

6. Fix array with TAPE!!!

   - Centrifuge 5 minutes at 900 x g and recover sample flow-through for analysis.

   - Discard water of the used balance-plate; store the Sephadex plates.

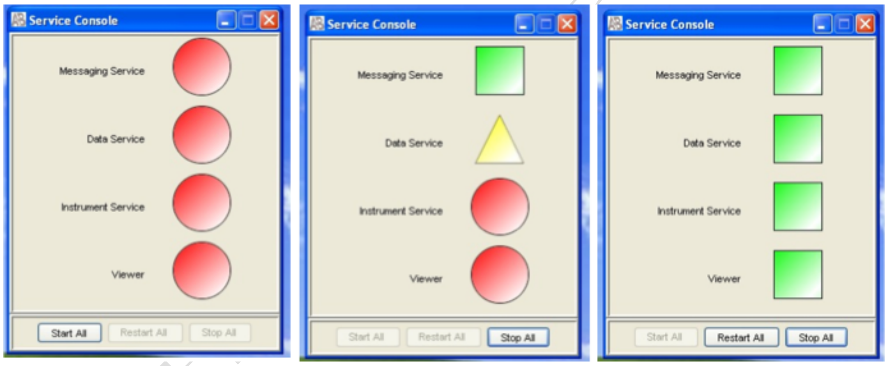
8. Load sequencer or store plate at +4°C, covered with saran wrap.

   - If you do not fill up sets of 4 wells, you need to add H_2O to missing wells for completing sets of 4 for the 4-capillary machine!. Otherwise capillary will be damaged if it aspirates air!!

SEQUENCER LOADING

START DATA COLLECTION SOFTWARE

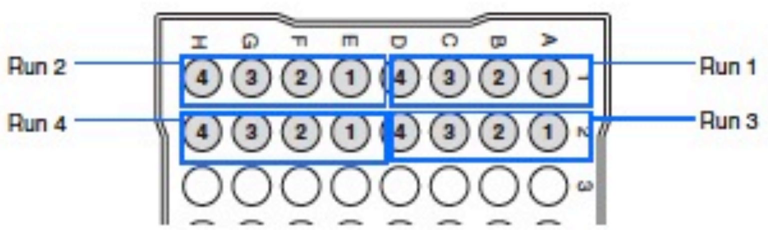
- Double click on **Run 3130 Data Collection v3.0** to display the Service Console. By default, all applications are off, indicated by the red circles. They launch automatically with the 3130 Data Collection software.
- As each application activates, the red circles (off) change to yellow triangles (activating), and then to green squares (on) when they are fully functional.



LOADING SAMPLES

Do not use distorted or damaged plates.

- For 3130 genetic analyzer, always fill 4 wells with your samples; otherwise fill with water (20µL / well) as illustrated below.



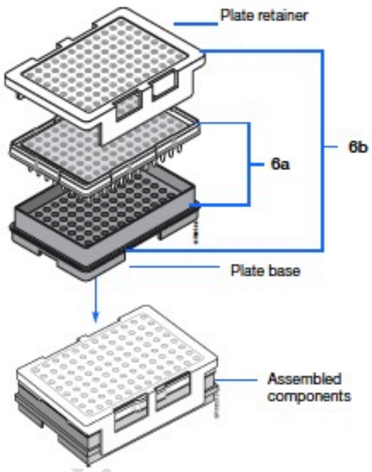
SEALING THE PLATE

- Place the plate on a clean, level surface.
- Place the septa flat onto the plate.

- Align the holes in the septa strip with the wells of the plate, then firmly press downward onto the plate. **Do not heat plates that are sealed with septa**.
- To prevent damage of the capillary array, inspect plate and septa to verify the septa fits snugly and flush on the plate.
- If the reagents in any well contain bubbles or liquid is not completely at the bottom of the well, briefly centrifuge the plate.
- Prepare the plate assembly and place the assembly on the autosampler, or store the plate at + 4°C.

ASSEMBLE THE PLATE ASSEMBLY

- As illustrated below, place the sample plate into the plate base.
- Snap the plate retainer onto the plate and plate base.



- Verify that holes of plate retainer and septa strip are aligned. If not, re-assemble the plate assembly. **Damage of the array tips will occur if plate retainer and septa strip holes do not align correctly.**

PLACING THE PLATE ASSEMBLY IN THE INSTRUMENT

- Verify the oven and front doors are closed.
- Press the Tray button and wait for the autosampler to stop at the forward position; Open front doors.
- Place the plate assembly on the autosampler as illustrated below. There is only one orientation for the plate, with the notched end of the plate base pointing away from you. Close instrument doors.

STARTING A RUN

- In the toolbar of the Data Collection software window, click on the green arrow to begin the run.
- The Processing Plates dialog box opens. Click OK.

  Note: The instrument may pause before running the plate to raise the oven temperature.

2. **PRIMER SEQUENCES** (CAN BE ADAPTED, ACCORDING TO HIV SUBTYPE)

Name	Sequence
F-1818	AGA AGA AAT GAT GAC AGC ATG TCA GGG AGT
R-RT-NcoI4362	GCT GAC ATT TAT CAC AGC TGG CTA CTA TYT C
F-2005	CAG GGC CCC TAG GAA AAA GGG CTG TT
R-4303	TCA CTA GCC ATG GCT CTC CAR TT
F-2135	TCA GAG CAG ACC AGA GCC AAC AGC CCC A
F-2610	GTT AAA CAA TGG CCA TTG ACA GAA GAA A
R-2605	GGG CCA TCC ATT CCT GG
R-3010	CCA TCC CTG TGG AAG CAC ATT G
R-3584	TGG CTC TTG ATA AAT TTG ATA TGT CCA TTG